

OSHA NEAR-MISS INCIDENT REPORT #NM-2026-089

Department: Chemical Processing & Synthesis

Location: Reactor Bay 4 - Nitric Acid Dosing Skid

Equipment Involved: P-204 High-Pressure Metering Pump

Reporter Role: Chemical Process Engineer

Immediate Action Taken: Emergency shutdown button triggered, acid line depressurized

Incident Description and Narrative:

During morning line pressure testing on Reactor 4, observed intermittent acid spraying from the PTFE flange gasket on metering pump P-204. The flange spray shield had been improperly secured with a temporary zip-tie rather than the mandatory steel locking collar. Nitric acid droplets splashed onto the chemical apron sleeve of technician. If dosing pressure had reached normal 60 PSI, direct facial contact with concentrated HNO_3 would have occurred.

Investigation Findings and Safeguard Assessment:

Physical inspection confirms barrier defect and procedural violation. Immediate review of lockout/tagout and secondary containment is mandated before resuming operations.